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Dissertation/Project Coversheet

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1. Introduction

Understanding Bitcoin's asset character is fundamental to portfolio management; a question that has gained significance following the 2024 approval of spot Bitcoin Exchange Traded Funds (ETFs). Bitcoin volatility dynamics, characterised at high frequencies and between regimes, is the foundational statistic for determining its character, and in turn its portfolio position.

This study is contextualised within Tang and Xiong's (2012) “asset financialisation" framework. When assets are bundled into major investment indices, institutional rebalancing, rather than their own supply and demand, drives their prices to co-move with broader markets, transmitting market-wide volatility and macroeconomic shocks into their dynamics. Such assets lose the independent behaviour that made them useful diversifiers, their risks increasingly reflecting global sentiment over fundamentals.

By integrating Bitcoin into conventional financial markets, it may adopt the same portfolio rebalancing dynamics observed in commodity markets by Tang and Xiong (2012). This study, therefore, examines whether Bitcoin has become more systematically determined, more sensitive to macroeconomic innovations, and structurally different from its pre-ETF behaviour. Most importantly, this study focuses on volatility dynamics opposed to return dynamics.

2. Literature Review

2.1 Bitcoin's evolving asset character

A key reference point in Bitcoin’s hedging literature is Baur and McDermott’s (2010) distinction between a hedge and a diversifier. A true hedge must exhibit negative correlation during market stress, whereas low or zero average correlation supports only diversification benefits. This distinction matters because diversification often fails precisely during downturns, when it is most required.

Early studies generally suggested hedge-like properties. Dyhrberg (2016a) found Bitcoin largely uncorrelated with the FTSE index and inconsistently related to dollar exchange rates, implying hedging potential. Selmi et al. (2018), using Quantile-on-Quantile regression, identified negative correlations between Bitcoin and oil prices at high quantiles. Both, however, examine Bitcoin's speculative, illiquid, pre-institutional phase, and subsequent gains in liquidity and institutionalisation have likely transformed these properties.

Hedge-indicative results are often inconclusive when considered alongside Bitcoin's historically fat-tailed volatility. There are numerous studies (Bhuiyan et al. 2021; Bouri et al. 2017, 2018; Kang et al. 2020; Xu and Kinkyo, 2023) that use non-linear econometrics to determine hedge or safe-haven performance. The general conclusions are complex: that Bitcoin's hedging capabilities occur at short horizons and are quantile and regime dependent rather than structurally persistent. However, at extreme quantiles a model can mistake jumps, un-modelled breaks and outliers (all naturally high in Bitcoin) for permanent economic relationships, leaving it unclear whether these frameworks uncover a genuine safe-haven mechanism or merely the noise of speculative bubbles.

Conlon et al. (2020) and Klein et al. (2018) measure hedging behaviour during COVID-19 and general market downturns, respectively. Moving beyond statistical dependence to multi-asset portfolio modelling, they find that Bitcoin inclusion amplifies portfolio downside risk through volatility spillovers during downturns; a finding that return-based studies haven't structurally captured.

The literature regarding Bitcoin's regime-specific post-ETF dynamics, especially volatility dynamics, is sparse. Under Tang and Xiong's (2012) framework, ETF-approval predicts specifically that Bitcoin's volatility should exhibit increased co-movement with financial markets and heightened macro sensitivity, yet empirical verification remains inadequate. Published studies, Mehdian et al. (2025) and Babalos et al. (2025), identify potential evidence of financialisation from improved returns and cross-market volatility linkages. However, these studies concern the impact around the announcement date. Existing pre-prints, Krause (2025) and Hong et al. (2025), attempt to verify long-term institutionalisation behaviour with short-term data. Methodologically rigorous research with an extended sample period is required to determine financialisation.

2.2 Macroeconomic Factor Ecosystem

Krause (2025) identifies the importance of considering factor-ecosystem integration when assessing whether an asset has institutionalised, emphasising traditional equity risk premia such as Market, Size and Momentum. However, Tang and Xiong's (2012) framework specifically implicates macro-driven rebalancing shocks, not equity risk premia, as the primary transmission channel of financialisation, an alternative factor ecosystem.

Foundational studies by Chen et al. (1986) and Schwert (1989) show that macroeconomic variables such as industrial production, term spreads, and risk premia significantly influence

equity returns and volatility. Basak and Pavlova (2016) and Cheng and Xiong (2014) extend this logic to financialised assets, proposing that portfolio rebalancing by financial intermediaries introduces systematic volatility into previously segmented markets.

There is academic precedent for considering certain macroeconomic variables to determine asset character. The Dollar Index and investment-Market Uncertainty are the two most common, appearing in most of the previous sections' papers as DXY and VIX. Equity markets often display asymmetric sensitivity to VIX shocks (Aboura and Wagner, 2016; Bekaert and Hoerova, 2014), suggesting that a more institutionalised Bitcoin should exhibit similar behaviour. Likewise, Bitcoin's historically negative sensitivity to DXY, associated with its anti-fiat narrative, may weaken if institutional ownership increasingly dominates price formation.

Interest rates and real yields are also widely used in asset-classification studies, especially for gold. Erb and Harvey (2025) show gold's rate sensitivity shifted materially after ETF-driven financialisaton; a direct analogue if Bitcoin has similarly integrated.

Gill-de-Albornoz et al. (2024) also find that Economic Policy Uncertainty (EPU) positively affects Bitcoin, interpreting this as evidence of investment-asset characteristics. Critically, VIX (market uncertainty) is a more appropriate investor sentiment measure. Their results may equally be characteristic of a speculative retail asset sensitive to a different type of uncertainty. A broader multi-factor framework is therefore preferable for assessing institutionalisation.

2.3 History of Bitcoin's Volatility Dynamics

Dyhrberg (2016b), using GARCH-type models, suggested that Bitcoin shared characteristics with gold and currencies without fully resembling either. Katsiampa (2017) found that the AR-CGARCH model, which separates short- and long-run volatility components, best captured Bitcoin volatility between 2010 and 2016. These findings imply that Bitcoin volatility may contain some fundamental value, inconsistent with purely speculative pricing.

A further nuance is that Bitcoin volatility may have a regime-dependent systematic component, rather than a long-run value. Later studies evidence regime-dependent volatility dynamics, including regime-dependent systematic components (Ardia et al. 2019; Chaim and Laurini, 2018; Shen et al. 2020). Importantly, the statistical "regimes" Ardia et al. (2019) and Shen et al. (2020) model are different concepts from an economic "regime" which refers to a global state, such as bear vs bull, or pre-ETF vs post-ETF. There could be several statistical structural breaks within a global state. This study adopts the economic interpretation of regimes, consistent with the post-Keynesian concept of non-ergodicity: relationships observed before ETF approval may

not reliably predict post-ETF behaviour because the institutional structure generating those relationships has changed.

Iyer and Popescu (2023) consider a "risk-off" regime characterised by periods of macroeconomic contraction, liquidity squeezes, or broad geopolitical shocks. They find increasing volatility linkages between cryptocurrencies and traditional financial markets. This is consistent with non-ergodicity, but their use of daily data overlooks the high-frequency volatility dynamics increasingly relevant in algorithmic cryptocurrency markets. Additionally, their framework studies price relationships rather than the macroeconomic transmission channels underlying them. A more robust approach would be informative about the reasons behind the increase.

Lu and Fang (2026) address the high-frequency limitation by constructing daily volatility from minute-level data. They find significant volatility spillovers from NASDAQ and gold markets into Bitcoin, with these relationships strengthening after September 2022. They statistically identify the 11th of September as a regime break, which corresponds to a month of major monetary policy shocks and technological developments in cryptocurrency markets. Therefore, it is important to clarify that this economic regime study is not one of causal attribution but of regime-specific behaviour.

2.4 Hypothesis Development

Three gaps motivate this study's design: absent long-horizon post-ETF verification, reliance on models ill-suited to Bitcoin's persistence and jump structure, and no test of the rate-channel implied by Tang and Xiong (2012). Four hypotheses follow. The literature motivates four testable hypotheses of Bitcoin within the post-ETF regime:

H1: Bitcoin's realised volatility process is structurally simpler and more stable. If financialisation reduces the relative contribution of idiosyncratic speculative dynamics, one would expect a reduction in jump frequency, excess kurtosis, alongside improved within-regime model fit.

H2: Bitcoin volatility exhibits asymmetric sensitivity to equity market uncertainty shocks. The literature documents asymmetric VIX responses in traditional equity markets (Aboura and Wagner, 2016; Bekaert and Hoerova, 2014). If Bitcoin has institutionalised, positive uncertainty shocks, corresponding to risk-off episodes, should amplify realised volatility more than negative shocks attenuate it. The prior regime, dominated by retail-driven anti-fiat narratives, provides the counterfactual.

H3: Bitcoin volatility exhibits emergent sensitivity to real interest rate, and term spread innovations. Erb and Harvey's (2025) gold analogue demonstrates that rate sensitivity can emerge following ETF-driven financialisation. Under Basak and Pavlova's (2016) rebalancing mechanism, institutional portfolio adjustments plausibly sensitise Bitcoin volatility to rate channels.

H4: Bitcoin realised volatility loses sensitivity to macro-fiat narratives, dollar appreciation, inflation expectations, and policy uncertainty. These relationships, present in the prior regime, reflect Bitcoin's historical positioning as an anti-fiat speculative asset (Dyhrberg, 2016a, 2016b). Their weakening would be consistent with institutional ownership displacing retail-driven price formation.

3. Methodology

3.1 Data

This is a regime comparison: it measures how Bitcoin's realised-volatility dynamics differ between the pre- and post-ETF windows, using the January 2024 approval as a thematically motivated boundary. Observed differences are associations across regimes, not causal effects.

High-frequency Bitcoin price data were sourced from the Binance REST API using BTCUSDT spot Klines, selected for their superior liquidity, continuous 24/7 trading, and longer uninterrupted history relative to CME futures, making them suitable for high-frequency volatility estimation. Six daily macroeconomic series were also collected: the CBOE Volatility Index (VIX) and US Dollar Index (DXY) from Yahoo Finance, and the Treasury yield curve slope (T10Y3M), 10-year real yield (DFII10), 5-year breakeven inflation rate (T5YIE), and US Economic Policy Uncertainty Index (EPU) from FRED. Variable selection follows the discussion in Section **2.2**. All series span 1 January 2020 to 31 March 2026 and are divided into a pre-ETF period (1 January 2020–9 January 2024) and a post-ETF period (10 January 2024–31 March 2026), hereafter referred to as the “prior” and “posterior” regimes.

BTCUSDT Klines provide quote data at multiple intervals. This study uses 5-minute closing prices, yielding 656,751 raw observations over the 6.25-year sample. Because Binance limits requests to 1,000 observations, data were retrieved in sequential batches using the final timestamp of the previous batch. From these prices, normalised 5-minute returns $r_{t,j}$ were

computed, with a maximum of 288 intervals per day. The midnight intervals were excluded to avoid returns overlapping days. Daily Bitcoin Realised Variance is constructed as:

$$RV_t^d = \sum_{j=1}^{M} r_{t,j}^2 \qquad r_{t,j} = \log P_{t,j} - \log P_{t,j-1} \quad \textbf{(1)}$$

Days with fewer than 230 observations (80% coverage) were excluded to remove abnormal low-activity periods. The final realised variance series contains 2,281 daily observations, of which 1,468 belong to the prior regime and 812 to the posterior regime.

Five-minute sampling is the conventional benchmark for realised variance estimation and is difficult to outperform empirically (Liu et al. 2015). It is also standard in the Bitcoin realised variance literature (Lu and Fang, 2026; Shen et al. 2020; Qiu et al. 2021), facilitating comparison with prior studies.

To analyse sensitivity to macroeconomic shocks, T10Y3M, T5YIE, and DFII10 were transformed into first differences (Δ), while VIX, DXY, and EPU were log-differenced (r) to account for scale-invariance. These transformed series represent macroeconomic innovations. Potential multicollinearity, particularly between DFII10 and T10Y3M, was assessed using Variance Inflation Factor tests (**A11**), confirming that no innovation introduced significant linear dependence. DFII10 and T10Y3M are mechanistically related, but their innovations may influence RV_t^d through different channels.

Variance modelling uses all 2,281 continuous-trading days; factor regressions drop weekends because the macro series aren't defined then, leaving 1,468 (942 + 526). Tables therefore report N=1,469/812 for variance statistics and N=942/526 for factor regressions.

3.2 Variance Modelling

The baseline specification is Corsi's (2009) Heterogeneous Autoregressive model for Realised Variance (HAR-RV):

$$RV_t^d = \beta_0 + \beta_d RV_{t-1}^d + \beta_w RV_{t-1}^w + \beta_m RV_{t-1}^m + \varepsilon_t \quad \textbf{(2)}$$

HAR-RV assumes that realised variance is driven by short-, medium-, and long-run volatility components, typically represented by daily, weekly, and monthly realised variance. The weekly and monthly components are defined as:

$$RV_{t-1}^{w} = \frac{1}{7}\sum_{i=1}^{7} RV_{t-i-1}^{d}, \qquad RV_{t-2}^{m} = \frac{1}{30}\sum_{i=1}^{30} RV_{t-i-1}^{d} \quad \textbf{(3)}$$

HAR-RV has been shown to outperform GARCH-type models for realised variance forecasting. Corsi (2009) argues that volatility persistence arises from heterogeneous market participants operating across different horizons, a dynamic not fully captured by the exponentially decaying memory structure of standard GARCH models. This behaviour is relevant for Bitcoin, whose speculative trading activity may amplify cross-horizon effects. Bergsli et al. (2022) further show that HAR models outperform several GARCH specifications, including APARCH specification that corrects for memory-decay property. Their evaluation across multiple accuracy criteria, including MCS procedures, confirms HAR suitability for high-frequency data.

Some conventions of computing RV_{t-1}^{w} and RV_{t-1}^{m} assume 5-day weeks and 22-day months, consistent with trading calendars (Lu and Fang, 2026). However, because Bitcoin trades continuously, weekend volatility likely contributes to weekday variance dynamics. This is why this study uses the entire 2281 observations and **(3)** assumes a 7-day week and 30-day month.

Given Bitcoin's susceptibility to idiosyncratic spikes, models incorporating jump detection were also considered. Comparing jump and non-jump specifications across regimes provides insight into how the role of jumps in realised variance has evolved. The jump components are defined as:

$$J_{t,\alpha}(M) = \left[RV_{t,M} - BPV_{t,M}\right] \times \mathbb{1}[Z_{BPV}(t,M) > \Phi_{\alpha}] \quad \textbf{(4)}$$

$$BPV_t = \mu_1^{-2}\sum_{j=2}^{M}\left|r_{t,j-1}\right| \cdot \left|r_{t,j}\right|, \qquad \mu_1 = \sqrt{2/\pi} \quad \textbf{(5)}$$

$$Z_{BPV}(t,M) = \sqrt{M} \cdot \frac{\left(RV_{t,M} - BPV_{t,M}\right)\cdot RV_{t,M}^{-1}}{\left[\left(\mu_1^{-4} + 2\mu_1^{-2} - 5\right)\cdot \max\left\{1, TQ_{t,M}\cdot BPV_{t,M}^{-2}\right\}\right]^{1/2}} \quad \textbf{(6)}$$

$$TQ_{t,M} = M \cdot \mu_{4/3}^{-3} \sum_{j=3}^{M} \left|r_{t,j-2}\right|^{4/3} \left|r_{t,j-1}\right|^{4/3} \left|r_{t,j}\right|^{4/3}, \qquad \mu_{4/3} = 2^{2/3} \cdot \frac{\Gamma(7/6)}{\Gamma(1/2)} \quad \textbf{(7)}$$

The jump factor component $J_{t,\alpha}$. represents the influence of jumps for a given days' volatility, which can be zero or otherwise. Bi-Power Variance BPV_t captures the continuous component of volatility, and it converges to RV_t when it contains no jumps. For days with statistically significant spikes in RV_t^d, the test statistic ZJ_{BPV} activates the indicator function $\mathbb{1}$, which in turn returns a non-zero $J_{t,\alpha}$. Tri-power Quarticity $TQ_{t,M}$ represents measurement-uncertainty in the relative contribution of the Jump to volatility, ensuring accurate ZJ_{BPV} detection. The significance level is set at $\alpha = 0.999$ to detect jumps with high certainty.

3.3 Model Selection: HARQ-F-J

Shen et al. (2020) compare 18 HAR-RV specifications, identifying the most accurate across different forecasting horizons, for both in-sample and out-of-sample data. Since the RV_t^d represents the last day in the sample, the ideal model would be most accurate for a 1-day in-sample forecast; this study selects the 4 most accurate from Shen et al. (2020), and their jump-variants, bringing the total to 8 models (**A1**). The criteria for optimality are high coefficients of determination alongside relative goodness of fit, assessed through the Akaike and Bayesian information criterion. HARQ-F-J performs well across both regimes.

$$RV_t^d = \beta_0 + \left(\beta_1 + \beta_{RQd} \cdot \sqrt{RQ_{t-1}^d}\right) RV_{t-1}^d + \left(\beta_7 + \beta_{RQw} \cdot \sqrt{RQ_{t-1}^w}\right) RV_{t-1}^w + \left(\beta_{30} + \beta_{RQm} \cdot \sqrt{RQ_{t-1}^m}\right) RV_{t-1}^m + \beta_j J_{t-1}^d + \varepsilon_t \quad \textbf{(8)}$$

$$RQ_{t-1}^d = \frac{M}{3} \sum_{j=1}^{M} r_{t-1,j}^2 \quad \textbf{(9)}$$

$$RQ_{t-1}^w = \left(RQ_{t-2}^d + RQ_{t-3}^d + \cdots + RQ_{t-7}^d\right)/7 \quad \textbf{(10)}$$

$$RQ_{t-1}^m = \left(RQ_{t-2}^d + RQ_{t-3}^d + \cdots + RQ_{t-30}^d\right)/30 \quad \textbf{(11)}$$

Realised Quarticity (RQ) captures measurement-uncertainty (typically caused by microstructure noise), adjusting it through $\beta_{RQd} \cdot \sqrt{RQ_{t-1}^d}$. The J_{t-1}^d component is the same as

(4). To account for seasonal variation (Lu and Fang, 2026) in Bitcoin volatility, day-of-week (DoW) dummies were added.

$$RV_t^d = \beta_0 + \cdots + \beta_J J_{t-1}^d + \sum_{d=1}^{6} \gamma_d D_{d,t-1} + \varepsilon_t \quad \textbf{(12)}$$

γ_d are dummies and $D_{d,t-1}$ are the corresponding binary indicators for Monday through Saturday, with Sunday omitted as the reference category. (**Table 1**) shows γ_d are collectively = significant at the 10% and 1% levels in the prior and posterior regimes. Additional tests confirm that γ_d do not interfere with macroeconomic regressors (**A8, A9**).

Finally, structural break dummies were incorporated to capture regime-specific volatility shifts:

$$RV_t^d = \beta_0 + \cdots + \beta_J J_{t-1}^d + \sum_{d=1}^{6} \gamma_d D_{d,t-1} + \sum_{k=1}^{K} \delta_k B_{k,t-1} + \varepsilon_t \quad \textbf{(13)}$$

Break dates were detected with the Iterative Cumulative Sum of Squares (ICSS) procedure, suitable for short-term regime detection. Here, $B_{k,t-1}$ denotes a binary indicator' and δ_k the corresponding level shift. K denotes the total number of breaks identified across the sample (**Fig.1*)**. The Sansó κ_2 detection test was applied as a robustness check, detecting no breaks in both regimes. ICSS likely over-identified breaks due to its sensitivity to high heteroskedasticity. Nonetheless, this study includes the breaks for variance-partitioning and improving forecasting accuracy (**A10**; Shen et al. 2020), not as economically interpreted events.

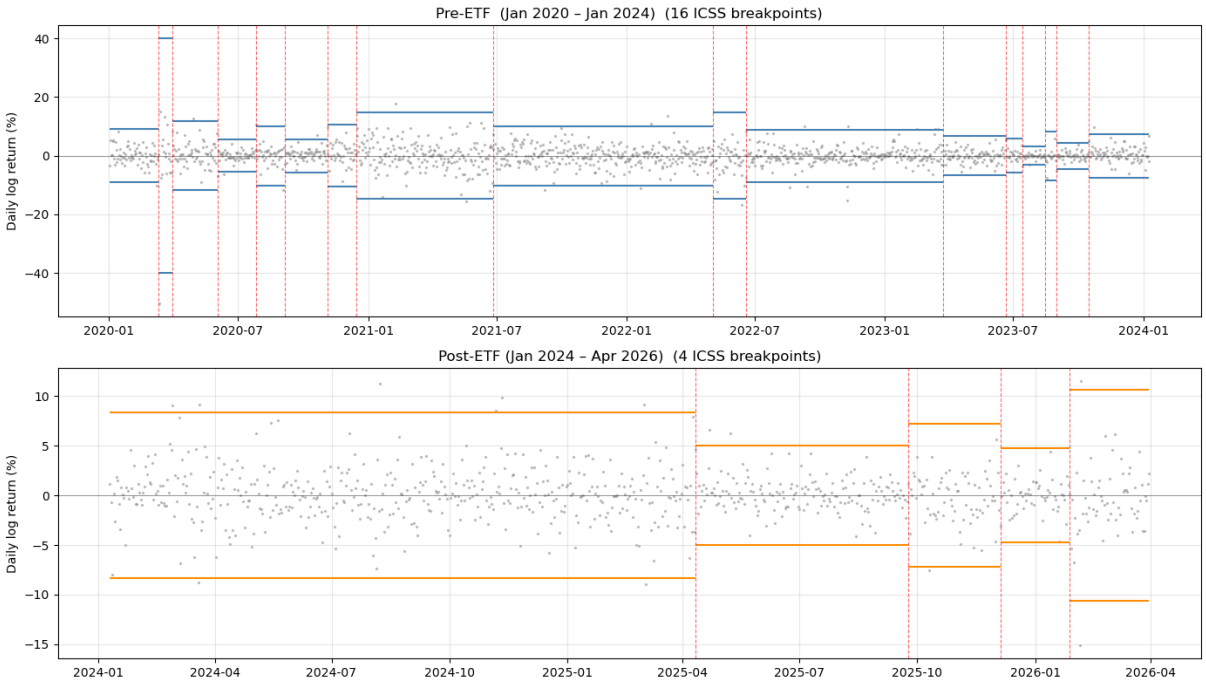


Figure 1: Bitcoin's daily log returns for pre-ETF and post-ETF samples with Iterative Cumulative Sum of Squares (ICSS) break points; yellow and blue lines encapsulate magnitude of returns.

Table 1. Overall Explanatory Power: With vs. Without Day-of-Week Controls

*The premium is the gain in HARQ-F-J adjusted R^2 attributable solely to the seasonal component. The last columns report the collective significance (F-test). *, **, *** indicate significance at 10%, 5%, and 1% respectively.*

Subperiod	Adj. R^2 Without Controls	Adj. R^2 With Controls	Premium	Collective Significance Probability Value
Pre-ETF (n = 941)	0.399	0.399	+0.000	0.08986*
Post-ETF (n = 525)	0.268	0.282	+0.014	0.00376***

3.4 Factor Regression

The baseline regression which incorporates the 6 factors $X_{m,t-1}$:

$$RV_t^d = \beta_0 + \cdots + \beta_J J_{t-1}^d + \sum_{d=1}^{6} \gamma_d D_{d,t-1} + \sum_{k=1}^{K} \delta_k B_{k,t-1} + \sum_{m=1}^{6} \phi_m X_{m,t-1} + \varepsilon_t \quad \textbf{(14)}$$

, where $m \in$ {ΔDFII10, rVIX, rDXY, ΔT10Y3M, ΔT5YIE, rEPU}. The final regression consists of asymmetric rVIX:

$$RV_t^d = \beta_0 + \cdots + \beta_J J_{t-1}^d + \sum_{d=1}^{6} \gamma_d D_{d,t-1} + \sum_{k=1}^{K} \delta_k B_{k,t-1} + \phi^+ rVIX_{t-1}^+ + \phi^- rVIX_{t-1}^- + \sum_{m=1}^{5} \phi_m X_{m,t-1} + \varepsilon_t \quad \textbf{(15)}$$

, where $rVIX_{t-1}^+ = \max(rVIX_{t-1}, 0)$ and $rVIX_{t-1}^- = \min(rVIX_{t-1}, 0)$ are the respective positive and negative components of $rVIX_{t-1}$. This is done considering the statistically different sensitivities the components have according to the Wald test (**A4, A5**). The decomposition will also allow me to identify the volatility asymmetry discussed in **Section (2.2)**.

Local projections (Jordà, 2005) are implemented ΔT10Y3M and ΔDFII10. If financialised, Bitcoin volatility would likely respond to these interest rate channels with a short lag, in line with institutional portfolio adjustments, as discussed in **Section (2.2)**. Rate signals are observed, positions are adjusted, and the effect manifests the next trading day. This transmission mechanism would be invisible to a same day ϕ_m.

Hence, for $x_{t-1} \in \Delta DFII10_{t-1}$, $\Delta T10Y3M_{t-1}$ and horizon $h = 0,1,2,3,5,10,$ the following regression is estimated separately:

$$RV_{t+h}^d = \alpha^{(h)} + \beta^{(h)} x_{t-1} + \sum_{l=1}^{L} \psi_l^{(h)} RV_{t-l-1}^d + \sum_{k=1}^{K} \delta_k^{(h)} B_{k,t-1} + \sum_{d=1}^{6} \gamma_d^{(h)} D_{d,t-1} + u_{t+h} \quad \textbf{(16)}$$

Which is essentially **(14)** augmented with $\beta^{(h)}$, which traces the response of RV_{t+h}^d to a unit shock in x_{t-1} at horizon h.

Because each regime is estimated separately, cross-regime differences cannot be established by comparing point estimates alone. The differences the study interprets, daily and weekly persistence, and the asymmetric VIX response, are therefore tested jointly in a pooled specification that interacts a post-ETF dummy with the HARQ-F-J cascade and macro terms,

with a Wald test on the interactions under HAC standard errors. HAC inference is used in place of a classical Chow test, which would assume equal error variance across regimes, an assumption the volatility results directly contradict

3.5 Quantile Regression Averaging (QRA)

This study concludes with Quantile Regression Averaging (QRA), allowing the relationship between RV_t^d and $rVIX_{t-1}$ to vary across the conditional distribution of RV_t^d. The response to $rVIX_{t-1}$ during calm, low-volatility regimes may differ from the response during elevated or extreme volatility episodes. The importance is abstracted from **Section 2.1**. A risk manager sizing a Bitcoin allocation needs to know not just its average sensitivity to risk-off shocks, but whether it amplifies when RV_t^d is already high. Implementing Koenker and Bassett (1978) with **(15)**, the τ- th conditional quantile of RV_{t+h}^d is:

$$Q_\tau\left(RV_t^d \mid \mathcal{F}_t\right) = \beta_0(\tau) + \beta_{1,t-1}(\tau)RV_{t-2} + \beta_{2,t-1}(\tau)RV_{t-2,t-7} + \beta_{3,t-1}(\tau)RV_{t-2,t-30}$$

$$+\beta_J(\tau)J_{t-1}^d + \phi^+(\tau)rVIX_{t-1}^+ + \phi^-(\tau)rVIX_{t-1}^- + \sum_{m=1}^{M-1}\phi_m(\tau)X_{m,t-1} + \sum_{d=1}^{6}\gamma_d(\tau)D_{d,t-1} + \sum_{k=1}^{K}\delta_k(\tau)B_{k,t-1} \quad \textbf{(17)}$$

Note that throughout this study the probability values are Newey–West (HAC) values, not standard OLS. HAC values are more appropriate for series with high heteroscedasticity. The finding of contrasting HAC and OLS probability values (**A2**; **A3**) validates this intuition.

Each hypothesis maps to a specific test: H1 to the cross-regime summary statistics, jump and break counts, and residual diagnostics; H2 to the asymmetric-VIX decomposition, its Wald symmetry test, and the QRA; H3 to the rate-channel local projections; and H4 to the contemporaneous macro-fiat coefficients across regimes. The pooled interaction test (**Table 10**) additionally assesses whether the persistence and VIX differences are significant across regimes.

4. Results

4.1 Volatility Maturation

The measures of spread (**Table 2**), Skewness, Excess Kurtosis, Standard Deviation and annualised volatility, have all decreased significantly. Skewness reduced from 17.33 to 6.66,

and Excess Kurtosis reduced from 394.28 to 73.27, evidencing lower heteroscedasticity, consistent with Jump reductions (**Fig2**). Notably, the very low Jarque-Bera probability values indicate that whilst bitcoins' heteroscedasticity has reduced, it remains highly statistically significant, reflecting that RV_t^d hasn't completely transformed, but is likely in a transitionary period.

Table 2. Bitcoin Daily Realised Volatility Summary Statistics

The statistics are reported for the pre-ETF and post-ETF regimes. The Jarque-Bera probability value indicates whether the distributional is Normal. N = 1469 opposed to the HARQ-F-J 1460 as the internal forecast leaves out the last day.

Statistic	**Pre-ETF (N = 1,469)**	**Post-ETF (N = 812)**
Mean	1.471e-03	6.881e-04
Median	7.211e-04	4.792e-04
Standard Deviation	4.132e-03	8.659e-04
Minimum	8.260e-06	2.370e-05
Maximum	1.104e-01	1.314e-02
Skewness	17.33	6.65
Excess Kurtosis	394.28	73.27
Jarque-Bera probability value	0.000e+00	0.000e+00
Annualised volatility (%)	73.29	50.12

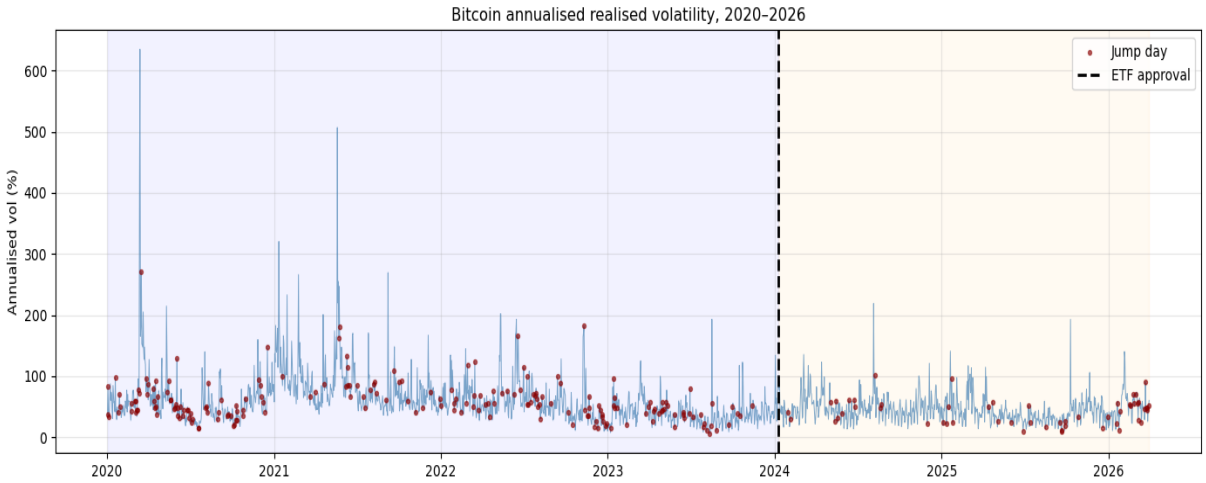


Figure 2: Time Series of Bitcoin's realised volatility over regimes. Red dots indicate jump days. Pre-ETF Jumps: 196 (13.5%), Post-ETF Jumps: 54 (6.7%). Dotted line marks ETF approval date.

Evidence of a simplifying volatility process can be found in the adjusted R^2 of the models, most of which reduced to approximately 16.5% (**Table 3**). The models' R^2 converging toward the same value indicates that model-specific extensions add less marginal value posterior. The HAR-RSV-J and HAR-PSL extensions, which had the highest prior R^2, are examples of this phenomenon. These extensions decompose RV_t^d into its positive and negative components to account for the leverage effect, which is known to cause fat tails and volatility clustering. Their relatively equitable posterior R^2 suggests weakening explanatory power of these properties.

Evidence of a stabilising RV_t is found in the ARCH and Ljung-Box (LB) tests which were initially statistically significant at less than 0.1% (**Table 3**), but almost all posterior tests were statistically insignificant, indicating an absence of residual and squared residual correlation. The general finding that posterior ARCH (5) yields a statistical significance of above 90% strongly suggests that posterior models, whilst having lower R^2 , better control heteroscedasticity. One interpretation consistent with this is a rising white-noise share and, tentatively, weaker short-horizon predictability; this cannot be established from these tests alone and is not pursued as a conclusion.

Note that the posterior HARQ-F-J remains significant (p=0.011) for LB (10), indicating the persistence of mild autocorrelation in its residuals, but as stated, most posterior tests are insignificant. Hence, stabilisation, if present, is broad but incomplete. To accurately identify internal dynamics, the HARQ-F-J coefficient estimates are examined.

Table 3. Model Adjusted R^2 Across Regimes (Snippet)

*Ljung-Box and ARCH-LM tests' probability values are also reported. *, **, *** indicate significance at 10%, 5%, and 1% respectively*

Model Specification	Pre-ETF Adj. R^2	Post-ETF Adj. R^2	Pre-ETF ARCH (5) Probability Value	Post-ETF ARCH (5) Probability Value	Pre-ETF LB (10) Probability Value	Post-ETF LB (10) Probability Value
HAR-PSL	0.454	0.165	0.000***	0.948	0.000***	0.134
HAR-RSV-J	0.448	0.166	0.000***	0.970	0.000***	0.104
HAR-ARJ	0.249	0.165	0.000***	0.976	0.000***	0.097*
HARQ-F-J	0.339	0.192	0.000***	0.999	0.000***	0.011**

The estimates (**Table 4**, **Table 5**) provide model-specific validation of a stabilising RV_t^d, increasingly characterised by white noise. Prior, two of the baseline HARQ-F-J coefficients are significant at 10%, whereas posterior, seven are significant, with five of them being at the 1% level. This is a dramatic shift in line with reduced autocorrelation and ARCH effects, potentially a consequence of a reduction in speculative trading behaviour.

Regarding white noise, recall that β_{RQh} corrects for measurement uncertainty. All three β_{RQh} increased in significance and magnitude, including β_{RQw} going from −1.42261 to −11.73086, and β_{RQm} going from −15.42795 to −40.01108. The main possible drivers are intraday kurtosis, jump activity, microstructure noise, or genuine white noise. The first two are ruled out empirically (5-minute excess kurtosis falls, **A12**; jump frequency decline, **Fig.2**). Further, institutional entry tightens spreads rather than widening them, making microstructure noise unlikely. A rising white-noise share is the most coherent remaining explanation. This, however, is offered as a candidate reading rather than established weak-form efficiency.

β_d reduces from 0.89486 to 0.37655, β_m reduces (in magnitude) from 0.18620 to −0.09188, and is statistically insignificant (p = 0.5490), and β_w rises from −0.07925 to 0.36456. These findings suggest that the very short-run and long-run memory components of volatility have weakened posterior, whilst the medium-run component (0.36456) has risen to closely match the short run (0.37655). However, these coefficients themselves aren't statistically distinguishable (**Table 10**), therefore the reallocation is described, not asserted as a structural shift in β_h.

An interpretation for improved β_w significance is that posterior volatility behaviour is increasingly driven by institutional participants operating on weekly cycles (trading calendars, weekly macro data releases, ETF rebalancing). This interpretation gains more merit when considering the finding that posterior weekly seasonality has become statistically significant (Lu and Fang, 2026).

These results support **H1**: the post-ETF process is simpler and more stable with reduced annualised volatility, jumps, breaks, excess kurtoFsis, and the loss of significant ARCH and Ljung–Box effects. Persistent non-normality and mild HARQ-F-J residual autocorrelation (LB p=0.011) mark this as a transition, not a completed reclassification. The fall in R^2 may be interpreted as evidence against **H1**, but it may also reflect lower unconditional variance (**A1**), not model deterioration.

Table 4. HARQ-F-J 1-Day in-sample Forecast: Pre-ETF Subperiod (N = 1469)

*All inference is based on HAC standard errors. *, **, *** indicate significance at 10%, 5%, and 1% respectively.*

Independent Variable	Coefficient	HAC Std. Error	Probability Value
Baseline Realised Volatility	0.00011	0.00032	0.718
Daily Persistence	0.89486	0.45584	0.050*
Noise Adjustment (Daily)	−2.69913	1.45833	0.064*
Weekly Persistence	−0.07925	0.45095	0.861
Noise Adjustment (Weekly)	−1.42261	2.91549	0.626
Monthly Persistence	0.18620	0.41909	0.657
Noise Adjustment (Monthly)	−15.42795	14.18021	0.277
Daily Jump Variation	−1.19832	0.80357	0.136

Table 5. HARQ-F-J Estimation Results: Post-ETF Subperiod (N = 812)

*All inference is based on HAC standard errors. *, **, *** indicate significance at 10%, 5%, and 1% respectively.*

Independent Variable	Coefficient	HAC Std. Error	Probability value
Baseline Realised Volatility	0.00041	0.00010	0.000***

Independent Variable	Coefficient	HAC Std. Error	Probability value
Daily Persistence	0.37655	0.06499	0.000***
Noise Adjustment (Daily)	−6.96514	1.71705	0.000***
Weekly Persistence	0.36456	0.10230	0.000***
Noise Adjustment (Weekly)	−11.73086	5.38878	0.029**
Monthly Persistence	−0.09188	0.15329	0.549
Noise Adjustment (Monthly)	−40.01108	21.89098	0.068*
Daily Jump Variation	−0.61326	0.18660	0.001***

4.2 Factor Regression Analysis

Prior, only positive shocks to market uncertainty were statistically significant at 5% (**Table 6**; **Table 7**), indicating RV_t^d increased with market uncertainty. Posterior, both positive and negative market uncertainty shocks are significant at the 10% level, with positive shocks still increasing RV_t^d, but negative shocks reducing RV_t^d. These findings are significant as they mirror the behaviour of traditional assets' volatility during risk-on and risk-off periods. Crucially, the finding that positive shocks have a greater magnitude of impact (0.00596, opposed to -0.00170), is consistent with the asymmetric hypothesis, a behaviour of traditional equities. Increasing market uncertainty has a greater influence on RV_t^d when it is already high, as demonstrated by the QRA (**Fig. 3**); the posterior RV_t^d is amplified (0.02166) at $\tau = 0.95$.

However, a joint Wald test (**Table 10**) cannot reject statistical equality of the daily-persistence, weekly-persistence, and asymmetric-VIX coefficients across regimes ($\chi^2(4) = 2.40$, $p = 0.66$). These channels are therefore best described as changes in point estimates rather than established structural shift. Hence, **H2** is only partially supported. Within the post-ETF regime, the VIX response is asymmetric, but the difference in the coefficients across regimes is not significant. Negative shocks are insignificant across the QRA, consistent with their very small mean coefficient (−0.00170): weak effects surface only when the mean regression averages across all days.

The prior (**Table 6**) sensitivities with the other innovations provide a valuable historical reference for RV_t^d evolution. Shocks in 5-year breakeven inflation were significant at 10% for

a negative relationship, indicating that spikes in expected inflation lowered Bitcoin volatility, potentially pointing toward its perception as an inflation diversifier during that regime. The shocks in economic policy uncertainty were significant at the 5% level, potentially reflective of Bitcoin's prior fragile regulatory landscape: policy shocks creating retail panic. It is important to not overstate these conclusions as these innovations' sensitivity coefficients are small (0.00032 and −0.00763). The statistical significance of the Dollar Index has more merit considering its large 0.13028 coefficient. Positive shocks to the dollar index significantly increased prior Bitcoin volatility. This reflects Bitcoin's historic role as an alternative "anti-fiat" currency, broadly consistent with the various hedging papers in **section (2.1).** H4 is supported: dollar, inflation and policy-uncertainty sensitivities present pre-ETF are absent post-ETF. The dollar channel is most compelling, with a change in magnitude rather than merely significance, which the smaller posterior sample cannot by itself explain. For inflation and policy uncertainty, whose coefficients were already small, the loss of significance is weaker evidence.

Notably, the absence of posterior significance for any of these factors mirrors the loss of unique price behaviours, induced by financialisation; when taken together with the posterior market uncertainty dynamics, the changes in RV_t^d are reminiscent of a retail asset sensitive to macro-fiat narratives (Dollar, Inflation, Policy Uncertainty), which, upon integration, possesses increased traditional asset sensitivities.

This analysis gains depth after interpreting multi-horizon rate-channel sensitivity. Prior Term Spread and TIPS yield shocks (**Table 8; Table 9)** indicate initial irresponsiveness to real rates. On the contrary, posterior shocks are significant at the one-day horizon. Precautionarily, their coefficients, 0.003615 and −0.003506, are small, however, both innovations are significant at the same horizon at 5%, even after seasonality and structural break controls. These results, considered alongside the factor and HARQ-F-J findings consistent with institutional participation, provide merit for the interpretation of that the rate sensitivity may be indicative of portfolio rebalancing. The institutional behaviour that would be observed under financialisation. This offers tentative support for **H3**: rate sensitivity appears post-ETF and is absent pre-ETF, but it is confined to the one-day horizon and rests on two of twenty-four horizon estimates. Given this multiplicity and the small coefficients, it is suggestive, not established.



Figure 3: Bar graphs of Quantile Regression Averaging on Market Uncertainty. X-axes represent RV_t^d quantiles τ. Y-axis represents RV_t^d sensitivity to market uncertainty at that τ. Key takeaway: tail amplification at τ=0.95 post-ETF.

Table 6. Macroeconomic Innovations Regression on Bitcoin Volatility: Pre-ETF Subperiod (942 obs.)

*HARQ-F-J regression with day-of-week and structural break dummies. Dummy coefficients are excluded. *, **, *** indicate significance at 10%, 5%, and 1% respectively*

Macro Innovations	Coefficient	HAC Std. Error	HAC Probability Value
Market Uncertainty Positive Shock (β^+)	0.00825	0.00404	0.041**
Market Uncertainty Negative Shock (β^-)	−0.01378	0.00937	0.141
Term Spread	−0.00482	0.00404	0.233
Dollar Index	0.13028	0.06859	0.058*

Macro Innovations	Coefficient	HAC Std. Error	HAC Probability Value
10-year TIPS yield	0.00418	0.00368	0.256
5-Year Breakeven Inflation	−0.00763	0.00449	0.089*
Policy Uncertainty	0.00032	0.00015	0.032**

Table 7. Macroeconomic Innovations Regression on Bitcoin Volatility: Post-ETF Subperiod (526 obs.)

*HARQ-F-J regression with day-of-week and structural break dummies. Dummy coefficients are excluded. *, **, *** indicate significance at 10%, 5%, and 1% respectively*

Macro Innovations	Coefficient	HAC Std. Error	HAC Probability Value
Market Uncertainty Positive Shock (β^+)	0.00596	0.00306	0.051*
Market Uncertainty Negative Shock (β^-)	−0.00170	0.00092	0.064*
Term Spread	0.00083	0.00146	0.569
Dollar Index	−0.00001	0.01038	0.999
10-year TIPS yield	−0.00267	0.00186	0.151
5-Year Breakeven Inflation	0.00278	0.00272	0.305
Policy Uncertainty	−0.00003	0.00005	0.621

Table 8. Local Projection: 10-year TIPS yield Innovations (ΔDFII10)

*Estimating the response of Bitcoin realised variance to 10-year real rate shock at different trading days ahead. Multi-horizon forecasting removes observations. Specification: HARQ-F-J with day-of-week and break dummies. Standard errors are HAC. Bold rows indicate significance at the 10% level or better. * p<0.10 ** p<0.05 *** p<0.01*

Horizon (h)	Coefficient (β)	Std. Error	p-value	Observations
Pre-ETF Subperiod				
0	−0.006267	0.004734	0.186	819
1	0.003679	0.004643	0.428	818
2	0.009039	0.007564	0.232	817
3	0.017226	0.012946	0.183	816
5	−0.008092	0.007799	0.299	814
10	−0.013961	0.009400	0.137	809
Post-ETF Subperiod				
0	0.000987	0.001659	0.552	447
1	**0.003615**	**0.001763**	**0.040****	**446**
2	−0.001909	0.003108	0.539	445
3	0.000848	0.001962	0.666	444
5	0.002833	0.002968	0.340	442
10	0.001591	0.001830	0.385	437

Table 9. Local Projection: Term Spread Innovations (ΔT10Y3M)

*Estimating the response of Bitcoin realised variance to Term Spread shock at different trading days ahead. Multi-horizon forecasting removes observations. Specification: HARQ-F-J with day-of-week and break dummies. Standard errors are HAC. Bold rows indicate significance at the 10% level or better. * p<0.10 ** p<0.05 *** p<0.0.*

Horizon	Coefficient (β)	Std. Error	p-value	Observations
Pre-ETF Subperiod				
0	0.005649	0.004334	0.192	819
1	0.001438	0.004477	0.748	818
2	−0.004309	0.004706	0.360	817
3	−0.009695	0.008698	0.265	816
5	0.007843	0.006383	0.219	814
10	0.011921	0.009030	0.187	809
Post-ETF Subperiod				
0	−0.003459	0.002320	0.136	447
1	**−0.003506**	**0.001663**	**0.035****	**446**
2	0.000755	0.002306	0.743	445
3	−0.002538	0.002131	0.234	444
5	−0.003698	0.003100	0.233	442
10	−0.001306	0.001474	0.376	437

Table 10. Cross-Regime Coefficient Equality Test

*HARQ-F-J coefficients augmented with asymmetric rVIX, the remaining macro innovations, day-of-week and structural break dummies. The final row reports the joint Wald test of all four differences. *, **, *** indicate significance at 10%, 5%, and 1% respectively.*

Coefficient	Pre-ETF	Post-ETF	Δ (post − pre)	HAC p-value
Daily persistence (β_d)	0.63435	0.66511	0.03075	0.898
Weekly persistence (β_w)	0.06227	0.34460	0.28233	0.471
VIX positive shock (φ^+)	0.00825	0.00596	−0.00229	0.652
VIX negative shock (φ^-)	−0.01379	−0.00167	0.01212	0.198
Joint Wald (H_0: all $\Delta = 0$)		**$\chi^2(4) = 2.398$**		**0.663**

4.3. Limitations and Recommendations

A few limitations should be kept in mind:

1)The sample is uneven across regimes, meaning estimates are more stable in the earlier regime and comparisons are not perfectly balanced.

2)Model choice is restricted. Only eight specifications from Shen et al. (2020) are considered, with the criteria of consideration being their study-dependent performance. HARQ-F-J is not necessarily the best possible model outside this framework.

3)There is also some disagreement over structural breaks. ICSS detects several, while the Sansó κ^2 test finds none. Break dummies are included mainly for theoretical and predictive reasons, since they improve fit (**A10**), but their statistical support is not fully consistent across tests.

4)The focus is on volatility, not return dynamics.

Future work should address these constraints directly. A broader model comparison, incorporating non-linear HAR extensions beyond those in Shen et al. (2020), or even HAR-alternative baselines, would strengthen model selection justification and provide insight into whether asymmetric amplification is an internal feature of Bitcoin Volatility. Consequently, alternative controls to ICSS may be identified, as well as long-term regime-break tests such as

Bai-Perron, which provide an alternative to an economically defined regime. As the post-ETF sample accumulates, replication with a longer and more balanced posterior window would improve cross-regime comparability and test power. Extending the analysis to return-based hedge regressions is the natural methodological complement to this study's volatility focus and would permit a complete asset classification assessment.

5. Conclusion

This study investigates the evolution of Bitcoin volatility and its sensitivity to macroeconomics factors in an increasingly institutionalised environment. The thematically motivated "breakpoint" for the institutional period was the US spot Bitcoin ETF January 2024 approval. Across a HARQ-F-J framework with horizon and quantile-varying macroeconomic augmentation from 2020-2026, the evidence is suggestive of a transition in volatility behaviour.

Annualised realised volatility falls from 73% to 50%, jump intensity drops from 13.5% to 6.7%. Standard diagnostic tests (ARCH and Ljung–Box), which strongly indicate heteroscedasticity and autocorrelation in the pre-ETF period, no longer do so post-ETF. At the same time, all model R^2 values converge to roughly 16–17% in the post-ETF period, suggesting a more stable and less heteroskedastic process overall. These findings corroborate **H1**, However, non-normality persists in both regimes, indicating that the change is better described as transitional rather than a full structural reclassification.

Within the HARQ-F-J specification, there is also a shift in volatility persistence. Pre-ETF dynamics are dominated by daily effects (β_d: 0.895 → 0.377), while post-ETF dynamics place equal weight on weekly components (β_w: −0.079 → 0.365). However, this cross-regime difference is not significant (**Table 10**); it is suggestive of, rather than evidence for, a move toward weekly institutional patterns.

The macroeconomic results provide partial support for financialisation. Pre-ETF sensitivities to the dollar index, inflation expectations, and policy uncertainty- often associated with speculative or narrative-driven trading, disappear after ETF approval, as **H4** theorizes. Post-ETF the market-uncertainty response is two-sided; positive shocks raise volatility, negative shocks dampen it, mirroring risk-on/risk-off equity behaviour, with upper-tail amplification (QRA, τ=0.95). This asymmetric response is consistent with behaviour observed in traditional equity markets, and with **H2.** Since the coefficients themselves aren't statistically different, **H2** is only subtly suggested. Quantile regressions further show that these effects are most

pronounced in the upper tail of the distribution, which is most relevant for risk management. Local projections additionally suggest that interest rate and term structure shocks become more relevant at short horizons post-ETF, consistent with institutional portfolio rebalancing effects hypothesis **H3**.

That said, the explanatory power of macro factors remains limited. The increase in fit is modest, and rate-based effects are statistically fragile. A cautious interpretation is that these linkages likely reflect correlated institutional flows rather than deep structural macroeconomic transmission. Overall, Bitcoin post-ETF is suggestively becoming a financialised risk asset: less volatile, more institutionally shaped, and more sensitive to market uncertainty, but still not fully integrated into macroeconomic fundamentals in the way that traditional equity markets are.

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A. Appendix

A1. In-Sample Adjusted R^2 Across Model Specifications

Adjusted R^2 for each HAR-family model specification estimated separately on the pre-ETF and post-ETF subperiods, without macro channels or seasonal controls. Δ reports the post-minus-pre change. The sharp drop in all specifications from pre- to post-ETF reflects lower unconditional variance in the post-ETF period rather than model deterioration. HARQ-F-J is the selected specification for the macro augmentation analysis.

Model Specification	Pre-ETF Adj. R^2	Post-ETF Adj. R^2	Δ (change)	Pre-ETF AIC	Pre-ETF BIC	Post-ETF AIC	Post-ETF BIC
HAR-RV	0.246	0.164	−0.082	−12062.936	−11957.516	−9275.702	−9238.116
HAR-RSV	0.446	0.165	−0.280	−12501.613	−12380.380	−9273.912	−9222.231
HAR-RSV-J	0.448	0.166	−0.282	−12506.771	−12380.267	−9273.636	−9217.256
HAR-PSL	0.454	0.165	−0.289	−12523.518	−12407.556	−9274.510	−9227.527
HAR-AR	0.247	0.166	−0.080	−12063.969	−11958.549	−9277.934	−9240.348
HAR-ARJ	0.249	0.165	−0.084	−12065.500	−11949.538	−9274.855	−9227.872
HARQ-F	0.335	0.191	−0.144	−12239.847	−12118.614	−9298.913	−9247.232
HARQ-F-J	**0.339**	**0.192**	**−0.147**	**−12247.461**	**−12120.957**	**−9299.133**	**−9242.754**

A2. ICSS Structural Break Dummies: Pre-ETF Subperiod

Binary indicators for variance break dates identified by the ICSS algorithm within the pre-ETF period. Each dummy takes the value 1 on its corresponding break date and 0 otherwise. Included to absorb abrupt level shifts in realised variance that would otherwise inflate residuals and distort HAR coefficient inference.

Independent Variable	Coefficient	OLS Std. Error	OLS p-value	HAC Std. Error	HAC p-value	OLS Misspecified at 5% (Yes/No)
Break Dummy 1 (2020-03-11)	0.01625	0.00132	0.000	0.01180	0.169	**Yes**
Break Dummy 2 (2020-03-30)	−0.01456	0.00119	0.000	0.01130	0.198	**Yes**
Break Dummy 3 (2020-06-02)	−0.00172	0.00067	0.010	0.00087	0.049	No
Break Dummy 4 (2020-07-26)	0.00014	0.00070	0.841	0.00022	0.525	No
Break Dummy 5 (2020-09-05)	−0.00021	0.00069	0.762	0.00017	0.213	No
Break Dummy 6 (2020-11-04)	0.00012	0.00070	0.860	0.00029	0.678	No
Break Dummy 7 (2020-12-15)	0.00111	0.00065	0.086	0.00062	0.075	No
Break Dummy 8 (2021-06-27)	−0.00105	0.00041	0.010	0.00060	0.079	**Yes**
Break Dummy 9 (2022-05-03)	0.00035	0.00054	0.519	0.00041	0.403	No
Break Dummy 10 (2022-06-19)	−0.00044	0.00055	0.422	0.00052	0.392	No
Break Dummy 11 (2023-03-24)	−0.00007	0.00042	0.859	0.00011	0.517	No
Break Dummy 12 (2023-06-21)	−0.00001	0.00079	0.991	0.00009	0.922	No

Independent Variable	Coefficient	OLS Std. Error	OLS p-value	HAC Std. Error	HAC p-value	OLS Misspecified at 5% (Yes/No)
Break Dummy 13 (2023-07-14)	−0.00005	0.00093	0.959	0.00010	0.637	No
Break Dummy 14 (2023-08-15)	0.00041	0.00104	0.692	0.00034	0.226	No
Break Dummy 15 (2023-08-31)	−0.00036	0.00099	0.719	0.00034	0.292	No
Break Dummy 16 (2023-10-15)	0.00006	0.00063	0.926	0.00014	0.679	No

A3. ICSS Structural Break Dummies: Post-ETF Subperiod

Same structure as Table 3. The substantially smaller number of breaks (4 versus 16 in the pre-ETF period) is consistent with reduced variance instability following ETF approval.

Independent Variable	Coefficient	OLS Std. Error	OLS p-value	HAC Std. Error	HAC p-value	OLS Misspecified at 5% (Yes/No)
Break Dummy 1 (2025-04-11)	−0.00029	0.00009	0.001	0.00007	0.000	No
Break Dummy 2 (2025-09-24)	0.00041	0.00013	0.001	0.00020	0.036	No
Break Dummy 3 (2025-12-05)	−0.00036	0.00016	0.022	0.00020	0.068	**Yes**
Break Dummy 4 (2026-01-28)	0.00037	0.00016	0.017	0.00013	0.005	No

A4. Directional Asymmetry Tests: Pre-ETF Subperiod (HARQ-F-J with seasonality and structural break dummy variables)

*Each macro channel is decomposed into positive and negative shock components and estimated within the full HARQ-F-J + breaks + Seasonality specification. The Wald F-test evaluates the null hypothesis that $\beta^+ = \beta^-$. A statistically significant Wald probability value warrants an asymmetric decomposition of the corresponding factor. *, **, *** represent significance at 10%, 5%, and 1% respectively.*

Macro Innovation	β^+ (Positive Shock)	β^+ Probability Value	β^- (Negative Shock)	β^- Probability Value	Wald F ($\beta^+ = \beta^-$)	p (Wald)
VIX Return	0.00760	0.071*	−0.01301	0.158	2.473	0.116
Term Spread	−0.00183	0.787	−0.00773	0.067*	0.583	0.445
Dollar Index Return	0.19080	0.068*	0.07000	0.182	1.596	0.206
10-year TIPS yield	0.00530	0.275	0.00355	0.328	0.187	0.666
5-Year Breakeven Inflation	−0.00575	0.288	−0.00875	0.072*	0.308	0.579
Economic Policy Uncertainty	0.00053	0.204	0.00011	0.643	0.482	0.488

A5. Directional Asymmetry Tests: Pre-ETF Subperiod (HARQ-F-J with seasonality and structural break dummy variables)

*Each macro channel is decomposed into positive and negative shock components and estimated within HARQ-F-J with structural break dummies and day-of-the-week dummies. The Wald F-test evaluates the null hypothesis that $\beta^+ = \beta^-$. A statistically significant Wald probability value warrants an asymmetric decomposition of the corresponding factor. Bold rows indicate a significance at the 10% level or better. * p<0.10 ** p<0.05 *** p<0.01.*

Macro Channel	β^+ (Positive Shock)	β^+ Probability Value	β^- (Negative Shock)	β^- Probability Value	Wald F ($\beta^+ = \beta^-$)	p (Wald)
VIX Return (symmetric)	**0.00609**	**0.069***	**−0.00178**	**0.151**	**3.113**	**0.078***
Term Spread	0.00195	0.276	0.00005	0.985	0.264	0.607
Dollar Index Return	−0.02310	0.271	0.01831	0.282	1.695	0.193
10-year TIPS yield	−0.00215	0.199	−0.00335	0.227	0.198	0.657
5-Year Breakeven Inflation	0.00231	0.186	0.00319	0.515	0.027	0.869
Economic Policy Uncertainty	0.00000	0.988	−0.00004	0.555	0.081	0.776

A6. Model Comparison: Symmetric vs. VIX-Asymmetric Macro Specification

Adjusted R^2 and parameter count for the two candidate macro specifications within the HARQ-F-J specification that includes structural breaks and Day-of-Week (DoW) dummies.

Subperiod	Macro Specification	Adj. R^2	Parameters (k)
Pre-ETF	Symmetric macro channels	0.399	36
Pre-ETF	VIX asymmetric only	0.410	37
Post-ETF	Symmetric macro channels	0.282	24
Post-ETF	VIX asymmetric only	0.335	25

A7. Level-Effect Regressions: 5-Day Moving Average of Rate Level

(HARQ-F-J with seasonality and structural break dummy variables)

Tests whether the prevailing level of interest rates, rather than daily innovations, explains Bitcoin realised variance. The 5-day moving average of the rate level is added alongside the full HARQ-F-J with structural break dummies and day-of-the-week dummies. No significant level effects are detected in either subperiod, confirming that Bitcoin RV responds to rate innovations rather than rate levels. HAC-robust standard errors.

Channel	Subperiod	Coefficient	Std. Error	p-value	Obs.	Adj. R^2
5-Day MA: Real Rate (DFII10)	Pre-ETF	−0.000205	0.000289	0.478	819	0.403
5-Day MA: Real Rate (DFII10)	Post-ETF	−0.000197	0.000233	0.398	447	0.287
5-Day MA: Term Spread (T10Y3)	Pre-ETF	−0.000236	0.000285	0.408	819	0.403
5-Day MA: Term Spread (T10Y3)	Post-ETF	−0.000083	0.000120	0.489	447	0.287

A8. Impact of Seasonal Controls on Incremental Explanatory Power: Pre-ETF Subperiod

*Each row reports the sequential marginal adjusted R^2 contribution of each macro channel with and without day-of-week (DoW) dummies. Differences between columns indicate whether seasonality interacts with or absorbs the explanatory power of individual macro channels. Bold rows indicate significance in at least one specification at the 10% level or better. *p<0.10 **p<0.05 ***p<0.01.*

Macro Innovation	Δ Adj. R^2 (without DoW)	HAC Probability Value	Δ Adj. R^2 (with DoW)	HAC Probability Value	Interpretation
Implied volatility shock (rVIX)	−0.00008	0.246	0.00000	0.248	No material seasonal interaction
10-Year real rate innovation (dDFII10)	0.00044	0.229	0.00043	0.258	Virtually unchanged
Dollar index shock (rDXY)	**0.00893**	**0.066***	**0.00897**	**0.064***	**Slight strengthening with seasonality**
Term spread (dT10Y3M)	−0.00060	0.646	−0.00066	0.988	Weak channel in both specifications
5-Year Breakeven inflation (dT5YIE)	**0.00262**	**0.128**	**0.00382**	**0.090***	**Stronger after seasonal controls**
Policy uncertainty (rEPU)	**0.00031**	**0.044****	**0.00040**	**0.037****	**Slight strengthening with seasonality**

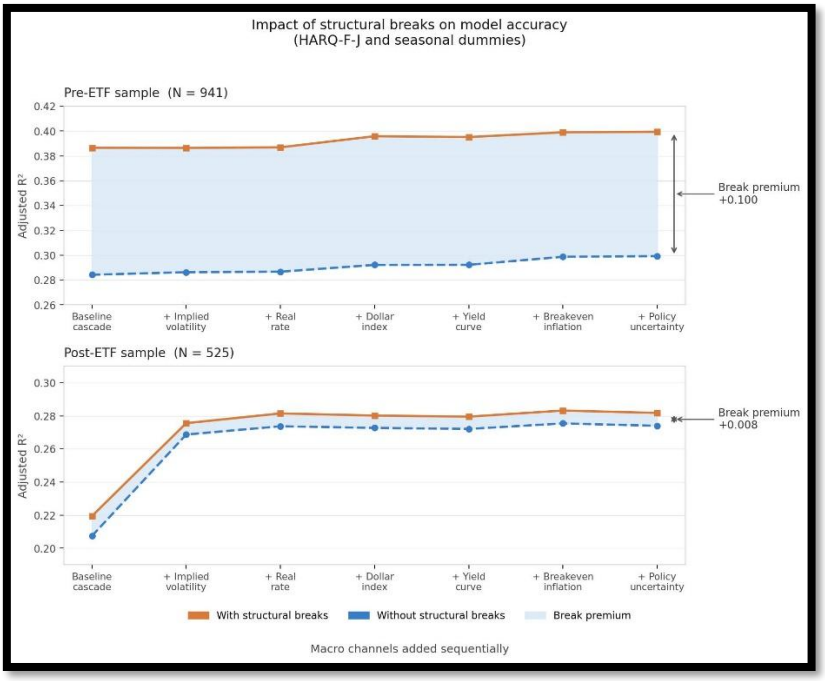
A9. Impact of Seasonal Controls on Incremental Explanatory Power: Post-ETF Subperiod

*Each row reports the sequential marginal adjusted R^2 contribution of each macro channel with and without day-of-week (DoW) dummies. Differences between columns indicate whether seasonality interacts with or absorbs the explanatory power of individual macro channels. Bold rows indicate significance in at least one specification at the 10% level or better. *p<0.10 ** p<0.05 *** p<0.01.*

Macro Innovation	Δ Adj. R^2 (without DoW)	HAC Probability Value	Δ Adj. R^2 (with DoW)	HAC Probability Value	Interpretation
Implied volatility (rVIX)	**0.05687**	**0.099***	**0.05609**	**0.094***	**Essentially unchanged**
10-Year real rate (dDFII10)	**0.00526**	**0.081***	**0.00589**	**0.069***	**Slight strengthening with seasonality**
Dollar index (rDXY)	−0.00142	0.908	−0.00129	0.771	No meaningful contribution
Term spread (dT10Y3M)	−0.00021	0.217	−0.00068	0.353	Weak in both specifications
5-Year Breakeven inflation (dT5YIE)	0.00403	0.317	0.00369	0.306	Broadly unchanged
Policy uncertainty (rEPU)	−0.00103	0.478	−0.00142	0.982	No meaningful contribution

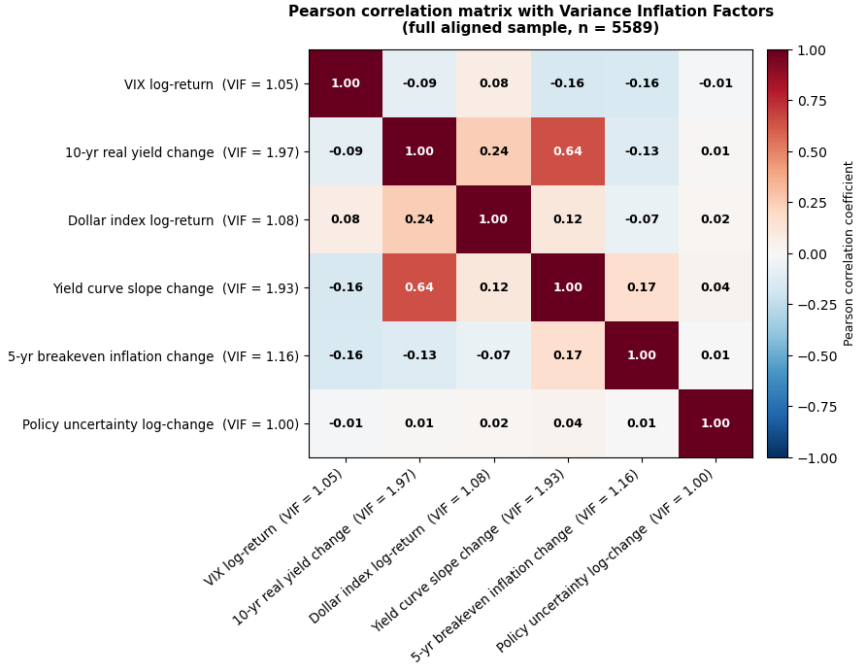
A10. Impact of Structural Breaks on Model Accuracy

Each data marker represents the sequential impact of the corresponding innovation in the x-axis. The shaded region represents the explanatory power contributed to the model by the inclusion of structural breaks on average across the sequence of factor-inclusions.



A11. Macro Innovation Correlation Heatmap

The overall Variance Inflation Factor (VIF) of each innovation is reported in brackets



A12. Bitcoin 5-minute Log-Returns Summary Statistics

Summary statistics of the 5-minute log-return series of Bitcoin. Fisher convention and the Pearson convention are both reported for excess kurtosis.

Statistics	**Pre-ETF (Observations: 421,207)**	**Post-ETF (Observations: 233,044)**
Standard Deviation	2.265e-03	1.548e-03
Skewness	0.03	-0.53
Excess kurtosis	182.50	49.95
Kurtosis (Pearson)	185.50	52.95